

Hot-Spot Architecture.

True FCFS for High-Contention Registration Systems

THE PROBLEM

A limited-inventory drop opens at peak hour.

30 million slots.

1 to 10 million requests per second at peak.

Users need real-time confirmation: did I get a slot?

CONSTRAINTS

True first-come-first-served · single counter can't scale ·
real-time response (no batch or queue results)

*The naive counter row collapses immediately —
it becomes a hot spot that serializes everything.*

What doesn't work.

✗ Single counter row

All traffic serializes on one row.

✗ Redis INCR/DECR alone

It's a cache — power loss risks claims.

✗ Message broker for atomicity

Crash before DB write loses the token.

✗ Two-phase commit

Theoretical. Nobody does it in practice.

THE INSIGHT

**The token store and the claim record
must live in the
same atomic transaction.**

Tokens ARE database rows. Claiming = one UPDATE in one tx.

Counters fight. Tokens don't.

THE SOLUTION

The architecture.

EDGE

Rate limiting · nginx / YARP token bucket



APP

hash(userId) → shard · power of two choices



DB

SELECT FOR UPDATE SKIP LOCKED + INSERT · fsync

```
BEGIN;

SELECT id FROM tokens
WHERE shard_id = @shard AND claimed_by IS NULL
LIMIT 1 FOR UPDATE SKIP LOCKED;  -- the magic

UPDATE tokens SET claimed_by = @userId,
                claimed_at = NOW(6) WHERE id = @tokenId;

INSERT INTO claims (user_id, token_id, ...) VALUES (...);

COMMIT;
```

WHY THIS WORKS

- Concurrent backends get DIFFERENT rows — no blocking
- Crash mid-tx → lock releases → token back in the pool
- Token loss impossible. Real atomicity. Real durability.

TL;DR

vs Redis Cluster.

	MySQL SKIP LOCKED	Redis Cluster
Atomicity	Native, one tx	Reconciliation
Durability	fsync per commit	AOF, 1s window
Token loss	Impossible	Possible on crash
Throughput	20–50k/s	100k–1M/s
Conceptual	Low — one model	High — many parts

TL;DR

- THE PATTERN Pre-allocation. 20+ years of industry practice.
- THE PRINCIPLE Turn one hot counter row into N cold token rows.
- THE PRIMITIVE SELECT ... FOR UPDATE SKIP LOCKED.

Counters fight. Tokens don't.

Conceptual integrity > buzzword throughput.